

Feature Recycling: Hiatus Resolution by way of Glide Formation in Japanese*

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Main Aims

- To provide a comprehensive description of glide formation as a productive hiatus-breaking strategy in Japanese.
 - To analyze the phenomenon from the viewpoint of the current phonological theory and consider some theoretical consequences. In brief:
 - It presents a formal analysis of autosegmental spreading in terms of Optimality Theory (Prince and Smolensky 1993) with recourse to Crisp Edge constraints (Ito and Mester 1994). Some extensions will be made to the original proposal.
 - It bears on the issue of the computation of alignment violations, in favor of the revision by Ito and Mester 1994 over the original proposal by McCarthy and Prince (1993).
 - It constitutes an instance of segmental phonology that crucially relies on syllabic structure in Japanese phonology.
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1. Data

1.1. Overview

The topic: the (optional) creation of glides homorganic to the first vowel in hiatus. The trigger can be [i], [e], [u] and [o]: the first two vowels create [j] and the other two create [w]. Only when these vowels are available are hiatal configurations broken up; hence the name *feature recycling*. The co-occurrence restriction of glides with a tautosyllabic vowel must be respected i.e., *wV[-low] (*wi,*we,*wu,*wo) and *jV[-back] (*je,*ji).

V1 \ V2	i	u	e	o	a
i	LongVowel	??	Phonotactics	[ijo]	[ija]
u	Diph/Phon	LongVowel	Diph/Phon	Diph/Phon	[uwa]
e	Diphthong	??	LongVowel	??	[eja]
o	Phonotactics	Phonotactics	Phonotactics	LongVowel	[owa]
a	Diph./Phon.	Diph/Phon.	Diph/Phon	Diph/Phon	LongVowel

The table above summarizes where glide formation takes place.

* Thanks are due to Takeru Honma, Kazutaka Kurisu, John McCarthy, Kohei Nishimura for their substantial help that I received in developing this work. Remaining errors are my own.

Parts represented by ?? are “weird gaps,” which will be discussed extensively below.

1.2. Concrete Data

1.2.1. Basic Patterns

[J]-Formation

/ia/ => [ija] (Katayama 1998; Kubozono 2002)

dai[j]a	‘diamond’
pi[j]ano	‘piano’
si[j]awase	‘happiness’
sai[j]aku	‘worst’
mi[j]ai	‘blind date’

/io/ => [ijo]

i[j]on	‘ion’
ohai[j]o	‘Ohio’
oni[j]on	‘onion’

/ea/ => [eja]

he[j]a	‘hair’
pe[j]a	‘pair’
ohe[j]a	‘O’hare’
e[j]akon	‘air conditioner’

[W]-Formation

/ua/ => [uwa]

gu[w]ai	‘condition’
taku[w]an	‘a kind of pickle’
karuu[w]a	‘kahlua’
hu[w]an	‘worry’
hu[w]antei	‘unstable’

/oa/=>[owa]

do[w]a	‘door’
ro[w]a	‘ROA’
o[w]asisu	‘oasis’
so[w]ara	‘soarer’
ko[w]ara	‘koala’

1.2.2. Some Gaps

The following three vowel sequences seem to resist this type of glide formation:

/eo/ => *[ejo]

*ore[j]o	‘[a name of cookie]’
*mete[j]o	‘meteor’
*ne[j]o	‘neo-’
*ne[j]on sain	‘neon’
*re[j]on	‘Leon’

/eu/ => *[eju]

*ke[j]u	‘rare’
*see[j]uchi	‘see lion’
*me[j]ue	‘older people’

/iu/ => ??[iju] (Poser 1986, 1988; Kubozono 2002)

/iu/ => [juu] *[iju]	‘say’
*umi[j]+ushi	‘a kind of a sea animal’

1.2.3. A further puzzle

Derived hiatus exhibits different patterns of hiatus resolution

Pattern 1 owa => oa => aa (cf., do[w]a)
 uwa => ua => aa (cf., taku[w]an)

chitto+wa	chittaa	‘little+TOPIC’
koto+wa	kotaa	‘fact+TOPIC’
soitsu+to+wa	soitsutaa	‘that guy+with+TOPIC’
koitsu+wa	koitsaa	‘this+TOPIC’
akaku+wa	akakaa	‘red+TOPIC’

Pattern 2 eba => ea => jaa (cf., he[j]a) (Miyara 1980; Poser 1986, 1988)
 ewa =>ea => jaa (cf., he[j]a)
 iwa => ia => jaa (cf., dai[j]a)

reba	rjaa	‘CONDITIONAL SUFFIX’
hake+ba	hakjaa	‘if throw up’
totte+wa	tottjaa	‘for+TOPIC’
kotodewa	kotodjaa	‘fact+COPULA+TOPIC’
kocchi+wa	kocchjaa	‘this side+TOPIC’

1.3. Other properties

Optionality: The glide formation is optional, though in spontaneous speech it is highly productive. For example, in careful slow speech, it is possible to pronounce [daia] instead of [daija] (Kubozono 2002).

Accent: Accent seems irrelevant. Both accented and unaccented vowels can trigger glide formation.

Accented Trigger Vowel	Unaccented Trigger Vowels
í[j]on	pi[j]ano
hé[j]a	óni[j]on
dó[w]a	hu[w]an
ó[w]asis	gu[w]ai

Locality: Glide formation is strictly local.

isao	*isa[j]o	‘[a male name]’
shita	*shita[j]u	‘respect’

Boundaries: Morphological and syntactic boundaries can inhibit the formation of glides:

Interference by Morphological Boundaries

*?jyuu[w]atsu	‘pressure’	(cf., karuu[w]a	‘kahlua’)
*haru natsu [w]aki huyu	‘spring, summer, autumn, winter’	(cf., karuu[w]a	‘kahlua’)
*umoo[w]atsume	‘wool hunting’	(cf., do[w]a	‘door’)
*?de[j]au	‘encounter’	(cf., he[j]a	‘hair’)
*?ke[j]agaru	‘kick and jump’	(cf., he[j]a	‘hair’)
*?mushi[j]atsui	‘humid’	(cf., pi[j]ano	‘piano’)
*mi[j]omo	‘pregnant’	(cf., ohai[j]o	‘Ohio’)

Interference by Syntactic Boundaries

*?sonkei siaeru [w]aite	‘the trustable partner’	(cf., hu[w]an	‘worry’)
*arito [w]arahuru	‘every kind of’	(cf., do[w]a	‘door’)
*?suzusii [j]asa	‘cool morning’	(cf., pi[j]ano	‘piano’)
*oite [j]aru	‘left’	(cf., he[j]a	‘hair’)
*akai [j]osara	‘red dish’	(cf., ohai[j]o	‘Ohio’)

- Some complex words allow glide formation across a putative morphological boundary e.g., *hu+[w]an* ‘worry’ and *gu+[w]ai* ‘condition’. For those cases, however, the morphological boundaries are likely to be synchronically inert.

2. Phonology or Phonetics?

An important question to be addressed regarding this phenomenon is whether the glides inserted are phonological or phonetic. The weak phonetic property of the glides at first blush suggests that they are just transitional elements (i.e., automatic articulatory effect of juxtaposing a sequence of two vowels). A proper phonetic experiment is thus called for (and in fact Kingston suggested several possible approaches (p.c. 2002)). Though the phonetic experiment is impossible at the present moment, I suggest that the phenomenon is actually phonological, based upon the following reasons.

2.1. Influence of morphological and syntactic boundaries

If the phenomenon were indeed nothing more than phonetic transitional elements, then morphological and syntactic boundary should not have any effect on glide formation.

2.2. Glide-Vowel Concurrence Restrictions Enforced

The concurrence restrictions of glides with tautosyllabic a vowel, motivated elsewhere in Japanese phonology, affect the insertion of glides under discussion.

*i[j]e	‘house’	*ki[j]eru	‘extinguish’
*chuu[w]i	‘attention’	*tsu[w]i	‘unintentionally’
*u[w]e	‘above’	*o[w]eru	‘finish’
*juu[w]utsu	‘melancholy’	*soo[w]utsu	‘manic-depressive psychosis’

The concurrence restrictions are likely to be phonological: the prohibition against [j] with [-back] vowels can be attributed to a segmental OCP effect (Leben 1973; McCarthy 1986; Christdas 1988; see also Alderete 1997; Myers 1997; Ito and Mester 1998 *inter alia*). Then it stands to reason that the glide formation is also phonological.

It might be that glides are in fact inserted, but our grammar of Japanese intervenes our perception. This point must also be verified in experimental studies.

2.3. Unrounded Back Vowels create rounded glides

Phonetically unrounded [u] can create a rounded glide. If the phenomenon is purely phonetic, how can we explain this fact?

taku[w]an	‘a kind of a pickle’
gu[w]ai	‘condition’
karuu[w]a	‘kahlua’

This [+round] behavior of [u] is parallel to the labialization of /h/ before [u] into [ɸ].

This is opaque; non-surface-apparent in the sense of McCarthy (1999). This, however, is likely to be attributed to the language-particular phonetic implementation component which takes as input the phonological representation of an utterance and produces as output the concrete specifications of phonetic representation.

2.4. The weird gaps ([eu] and [eo])

Why these gaps, if the glides are just transitional elements? A phonological explanation is called for (see below 3.2.2).

3. Analysis

3.1. Preliminary

3.1.1. Glide insertion as spreading

The insertion of the glides should be captured as the spreading of the place of articulation from the first vowel in hiatus to the following syllable. This on the other hand should not be conceived of as a mere epenthetic element: if the glides were epenthesis, why wouldn't they repair all onsetless syllables?

Onsetless Syllables: no epenthetic glides

a.oi	*[j]a.oi	*[w]a.oi	‘blue’
i.chi	*[j]i.chi	*[w]i.chi	‘one’
u.mi	*[j]u.mi	*[w]u.mi	‘see’
e	*[j]e	*[w]e	‘picture’
oo.kii	*[j]oo.kii	*[w]oo.kii	‘big’

Moreover, if the glide was inserted out of blue, how could we explain the fact that the glide inserted share almost all of the features with the preceding vowel?

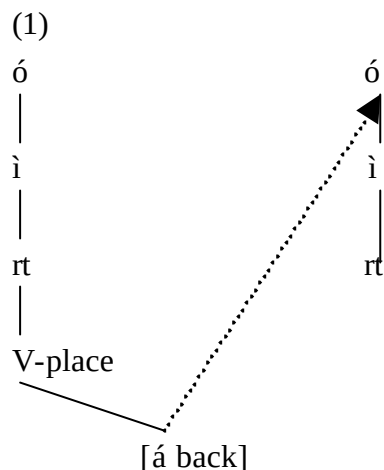
3.1.2. Theoretical Frameworks Employed

Feature Geometry Theory (Clements and Hume 1995 and reference therein): this is to formally capture the alternation as spreading. No particular version is assumed, however.

Moraic Theory (Hyman 1985; McCarthy and Prince 1986; Hayes 1989 and subsequent works): vocoids associated with a mora are interpreted as vowels; those that directly linked to syllable nodes are glides.

3.1.3. The formalization of the Alternation

In derivational terms, the rule can be formalized as the following rule:



The process is, in short, the spreading of [á back] (á being either plus or minus) of the first vowel to the second syllable. [high] is irrelevant: mid vowels can create high glides.

- Other necessary features are (i) either automatically interpolated (Sagey 1986) or (ii) implemented in the phonetic component.
- The rule is stated in its most general form. Other conditions block the application of the process (see below 3.2.2.)

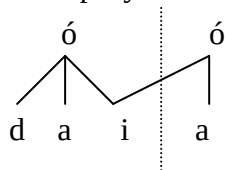
3.2. An OT analysis

3.2.1. Basic Ranking

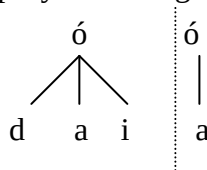
The trigger is Onset (Ito 1986, 1989; Prince and Smolensky 1993) because the alternation takes place to avoid hiatal configurations (or possibly No-Hiatus; McCarthy 1993; see Pullyblank 1998 ft 3; Orie and Pullyblank 1998 for discussion).

What is violated by the ambisyllabic parsing of a first vowel in hiatus is bi-partite. *SPREAD is a faithfulness constraint that prohibits a new autosegmental association. The other is a markedness constraint, Crisp Edge(ó) (Ito and Mester 1994; Merchant 1994):

Non Crisp Syllable Edge



Crisp Syllable Edge



The List of Relevant Constraints

- Onset: A syllable has a onset consonant
- *SPREAD: don't add a new association line (equivalent to Dep(association)).
- Crisp Edge (ó): A syllable edge must be crisp: all segments are uniquely syllabified.
- Max: No deletion (McCarthy and Prince 1995)
- Dep: No Insertion (McCarthy and Prince 1995)

No epenthesis or elision takes place in order to satisfy Onset in Japanese, hence Max and Dep must dominate Onset. However, ambisyllabic parsing takes effect to avoid hiatus, hence Onset >> CrispEdge(ó), *SPREAD.

Feature Recycling Ranking

Max, Dep >> Onset >> CrispEdge(ó), *SPREAD

/umi/	Max	Dep	Onset	CrispEdge(ó)	*SPREAD
a. [u.mi]			*		
b. [ʔu.mi]		*!			
c. [mi]	*!				

/daia/	Max	Dep	Onset	CrispEdge(ó)	*SPREAD
a. [dai.a]			*!		
b. [dai.ja]				*	*
c. [dai.ʔa]		*!			
d. [dai]	*!				

3.2.2. Blocking Constraints

Looking back at the chart in the first page, there are some circumstances where glide formation is blocked.

1. Why doesn't [a] create a glide?

High sonority of [a] is incompatible with consonantal parsing. Here this point is formally expressed as [a]=V, which prohibits nonmoraic parsing of [a] (Rosenthal 1994: 41, 1997: 156; McCarthy and Prince 1993b: 162-163 and reference cited therein).

2. How can we express the co-occurrence restrictions of glides with a tautosyllabic vowel?

The prohibition against [j] with [-back] vowel derives from the OCP (Leben 1973; McCarthy 1986; see also Rosenthal 1994: 198-202; Kurisu 2000: 171). Let us express the restriction of

[w] by means of an ad hoc constraint *wV[-low].

All of these constraints must dominate Onset:

/aoi/	[a]=V	Onset
a. [a.oi]		**
b. [a.oi]	*!	*

/kieru/	OCP	Onset
a. [ki.je.ru]	*!	
b. [ki.e.ru]		*

/koeru/	*wV[-low]	Onset
a. [ko.we.ru]	*!	
b. [ko.e.ru]		*

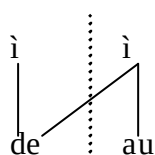
3. Morphological and Syntactic Boundary

Why does a morphological boundary interfere with the formation of glides?

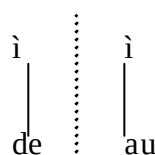
Proposal: CrispEdge constraints, first proposed by Ito and Mester for prosodic categories, are extended to morphological categories. i.e., CrispEdge (ì), ì representing a morpheme.

- CrispEdge (ì): every element in ì1 uniquely belongs to ì1 (assuming an autosegmental association of phonological elements with a morpheme (McCarthy 1979, 1981)).

Non Crisp Morpheme Edge



Crisp Morpheme Edge



CrispEdge (ì) dominates Onset:

/de+au/	CrispEdge(ì)	Onset
a. [de.jau]	*!	
b. [de.au]		*

4. How about syntactic boundaries?

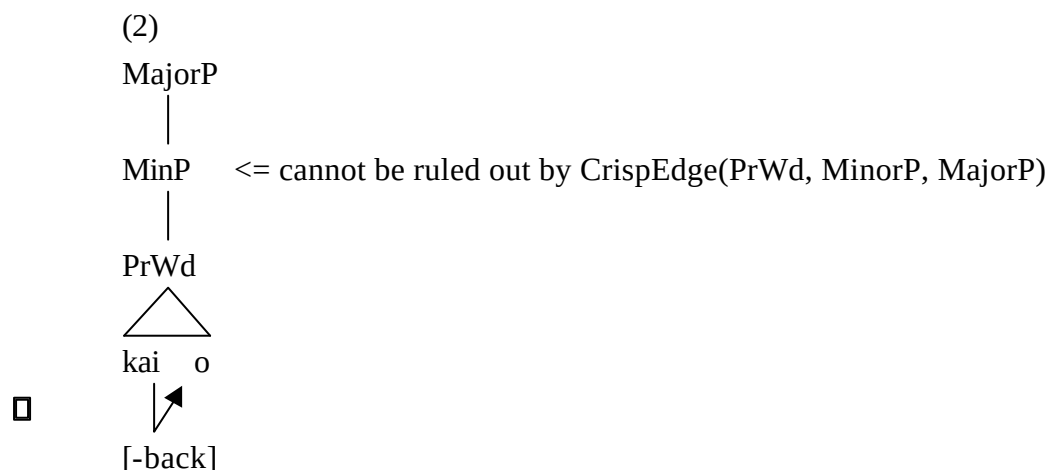
CrispEdge(i) takes care of the blocking by syntactic boundaries because syntactic boundaries necessarily entail the presence of a morphological boundary.

-An alternative might be to posit CrispEdge (XP), where XP represents a syntactic category. But, can OT constraints refer to syntactic information in this way? Obviously, the theory is more restrictive if phonology is blind to any syntactic information.

□

-How about CrispEdge constraints governing higher prosodic categories such as prosodic words, minor phrases or major phrases (Selkirk and Tateishi 1988; 1991 *inter alia*)? These are empirically inadequate. [i]-final nouns cannot spread its [i] when followed by the objective marker [o], which is assumed to be in the same prosodic word (and thus the same minor phrase) with the noun to which it attaches (Miyara 1980; Selkirk and Tateishi 1988, 1991).

□ *kai [j]o katta ‘bought a clam’ □
 □ *hachi [j]o katta ‘bought a pot



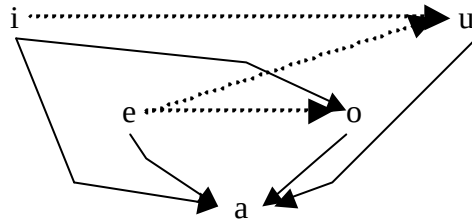
5. How about the gaps (i.e., [eo], [eu] and [iu])?

Recall that /eu/ and /eo/ seem to resist glide formation. /iu/ seems to undergo another type of glide formation i.e., [juu] (Poser 1986, 1988)

Ignoring /iu/ for the moment, these gaps are apparently weird in that:

- [ju] and [jo] are well-formed in Japanese e.g., *eejuu* ‘hero’ and *eejoo* ‘nutrition’.
- /ea/ becomes [eja] which indicates that /e/ is a legitimate trigger.

Hypothesis: none of /eu/, /eo/, /iu/ are hiatus but constitute a diphthong. Consider the vowel space now (solid lines represent the sequences where glide formation takes effect; dotted lines show the gaps):



The empirical generalization here is that the ambisyllabic parsing does not take place when the height of V1 is lower or the same as that of V2. In other words, the gaps arise when sonority falls from V1 to V2.

The fall in sonority might allow [eo], [eu] and [iu] to be a diphthong. Since they are in a same syllable, ambisyllabic parsing of V1 is intrinsically impossible.

One Future Question: is there other evidence that shows [eo], [eu] and [iu] are diphthongs?

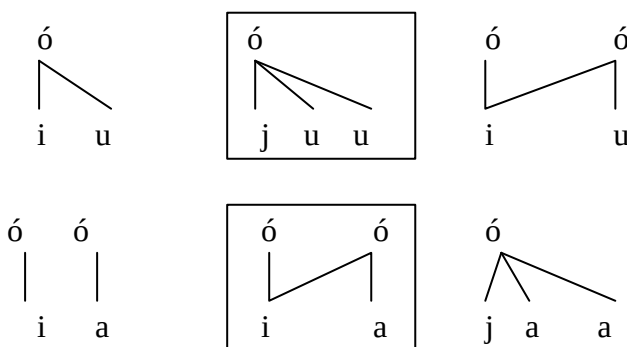
3.2.3. More Puzzles

1. How about the formations /iu/ => [juu] (Poser 1986, 1988)?

How is it that /iu/ becomes [juu] not *[iju], while /ia/ becomes [ija] rather than *[jaa].

From a purely segmental point of view, this asymmetry is hard to explain.

When we look at the syllable structure, a simple generalization emerges. The alternation always preserves syllabic affiliation of melodic elements, as illustrated below. (It can alternatively be that it is the number of syllables that is to be preserved).



Hence with a constraint such as Ident(ó), which militates against the change of syllabic affiliation, the contrast can be accounted for. Assuming that /iu/=> [juu] type glide formation incurs a violation of Ident(ì),

/iu/	Ident(ó)	Ident(ì)	*SPREAD
[juu]		*	
[iju]	*!		*
/ia/			
[ija]			*
[jaa]	*!	*	

One attendant problem: this approach hinges on the assumption that inputs have a syllable structure. This assumption is highly controversial. How to syllabify consecutive vowels is a language-particular matter, so assuming syllabification in input goes against the thesis of Richness of the Base (Prince and Smolensky 1993; Smolensky 1996; Kurisu 2000) – possible solutions; sympathy (cf., McCarthy’s (1999) treatment of Bedouin Arabic), serial OT, OO-Correspondence (Benua 1998) or OV(Output-Variant)-Correspondence (Kawahara 2001) etc.

Why [eu] and [eo] do not undergo any alternation remains mystery.

2. The Derived Environment Puzzle (1.2.3. on p3)

Recall that derived hiatus and underlying hiatus behave differently:

kore + wa => korea => korjaa	vs.	hea => heja
koitsu + wa => koitsua => koitsaa	vs.	huan => huwan

This, and in fact the problem of derived environment effect in general, raises a recalcitrant problem to the standard OT. There are two major proposals in current phonological theory concerning this issue: Local Conjunction (Lubowicz to appear) and Comparative Markedness (McCarthy 2002b). However, only the former approach works.

- Local Conjunction: the basic idea is that derived environments by their nature incur a violation of one faithfulness constraint – call it F1- and an otherwise lowest ranked constraint, conjoined with F1, is ranked higher (or “activated” as Lubowicz puts it). In our case at hand, the following ranking accounts for both derived and non-derived patterns.

{*SPREAD&Max} >> Onset >> Ident(i) >> *SPREAD

Non-Derived Environment

/hea/	{*SPREAD&Max}	Onset	Ident(ì)	*SPREAD
hea		*!		
heja				*
hjaa			*!*	

Derived Environment

/-reba/	{*SPREAD&Max}	Onset	Ident(ì)	*SPREAD
rea		*!		
reja	*!			*
rjaa			**	

-Comparative markedness theory recognizes two kinds of markedness constraints: Mo (M<old>) and Mn(M<new>). The former assigns a violation mark to any marked structure that fails to be eliminated in input-output mapping. Mn penalizes any new instance of the marked structure that is not present in the input (see McCarthy 2002b for comprehensive discussion and formalisms). Mn >> Faith >> Mo takes care of the standard derived environment effect, where only newly derived marked structures are eliminated. This does not, however, work for the Japanese case above: the repair strategy is unambiguously chosen on the basis of how the faithfulness constraints (*SPREAD and Ident(ì)) are ranked with respect to each other. In other words, no matter how you rank Mn and Mo, the repair strategy taken is determined depending upon which faithfulness constraint is ranked the lowest, so long as the target marked structure is identical.

4. Additional Theoretical Implications

4.1. Productive Onset-Driven Phenomenon in Japanese

As the chart in the first page suggests, Japanese actively eliminates hiatus when other conditions are met. This supports the thesis, which is admittedly already well-established in linguistic theory, that human languages prefer the presence of onset sounds (Ito 1986, 1989; Prince and Smolensky 1993), and Japanese is not an exception.

Japanese is replete with examples with onsetless syllables. However, the phenomenon we saw above suggests that the requirement that a syllable starts with a consonant is still active in the phonology.

This situation can be formally captured by the fundamental tenet of Optimality Theory (Prince and Smolensky 1993) that dominated constraints are not totally inactive (see also McCarthy and Prince 1993b, 1994; McCarthy 2002a). In other words, constraints are not like

parameters which can be only either on/off. Thus the phenomenon here provides another empirical support to Optimality Theory over a parametric theory.

4.2. The Status of Syllable Structures in Japanese

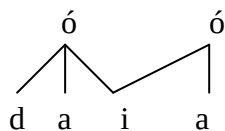
The role of syllables in Japanese is less eminent than that of moras. However, our treatment of the gaps discussed above crucially relies on the different syllable structures of the vowel sequences i.e., [ia] and [eo] must have a different syllable structure.

This provides another empirical endorsement to the thesis that syllables play a role in Japanese phonology, as previously argued by Vance (1987), Haraguchi (1996), Kubozono (1999) among others.

4.3. The formal property of alignment constraints

To the extent that the phenomenon is Onset-driven and Onset is formally expressed as an alignment (McCarthy and Prince 1993a; Ito and Mester 1994), then the phenomenon suggests that they are not violated in case of multiple linking.

Onset: Align-L (ó, C): the left edge of every syllable must be aligned with a consonant (McCarthy and Prince 1993a; Ito and Mester 1994).



Apparently, the left edge of the second syllable above does not seem properly aligned with a consonant due to the multiple linking of [i]. Empirically, however, the structure above must satisfy Align-L (ó, C).

This lends further empirical support to the premise that alignment proper is to be distinguished from crisp edge-ness, as first argued by Ito and Mester 1994 and later supported by Merchant 1994 and McCarthy and Cohn (1994). This is contra the analysis of Malay presented by McCarthy and Prince (1993a: 53-55) and Rosenthal (1994:182-190), who consider the multiple linking of ambisyllabic segments to cause a violation of alignment constraints.

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